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05/27/2026



Kria KR260 Robotics Starter Kit User Guide (UG1092)

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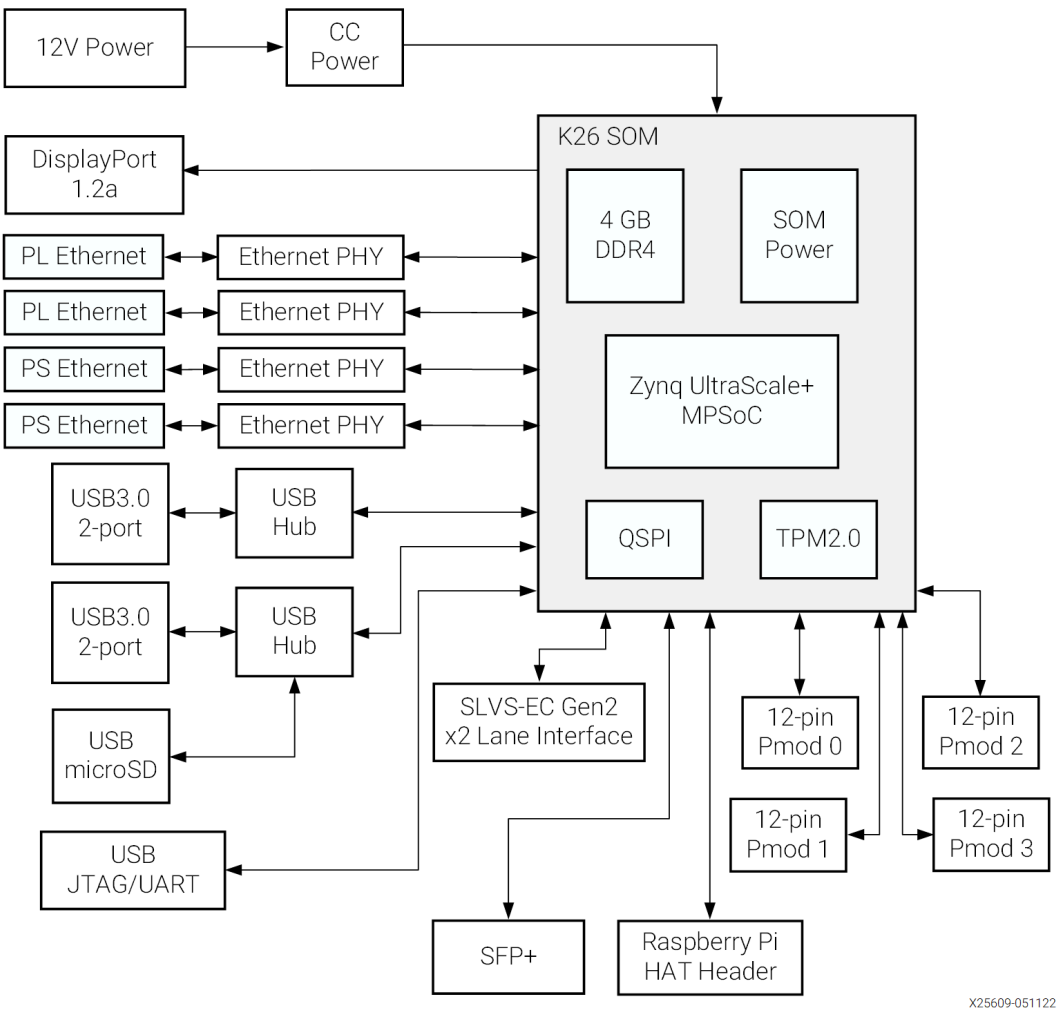
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Kria KR260 Robotics Starter Kit User Guide (UG1092)

Summary

The AMD Kria™ KR260 Robotics Starter Kit is comprised of a non-production version of the K26 system-on-module (SOM), carrier card, and thermal solution. The SOM integrates core digital hardware components including an AMD Zynq™ UltraScale+™ MPSoC, run-time memory, non-volatile boot devices, an integrated power solution, and a security module. The robotic-focused carrier card provides various application peripheral options including a sensor input, video display outputs, USB, SD card, Raspberry Pi HAT interface, Pmod headers, SFP+ connector, and Ethernet physical interfaces. The thermal solution includes a heat sink, heat sink cover, and fan. The Kria KR260 Robotics Starter Kit is designed to provide customers a platform to evaluate their target applications and ultimately design their own carrier card with K26 SOMs. Key target robotics applications are supported with an emphasis on industrial and automation markets.

Figure: KR260 Starter Kit Block Diagram



What's in the Box?

The Kria KR260 Robotics Starter Kit includes a K26 SOM, integrated thermal solution, and carrier card. The kit is only meant for SOM platform evaluation with the carrier card providing a variety of interfaces for integrating different peripherals. The Kria KR260 Robotics Starter Kit also includes the following accessories inside the box: power supply and its adapters, Ethernet cable, USB A–male to micro B cable, a microSD with adapter, and developer stickers. The box also includes a Getting Started card that directs you to the getting started web page and product page. This guide lists the [Supported Peripherals](#) that can be purchased separately.

Table: Summary of Box Contents

Line Item	Items	Quantity
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Line Item	Items	Quantity
1	Starter Kit (SOM with fansink thermal solution on top of robotic carrier card)	1
2	Getting Started card	1
3	Developer stickers	1
4	Power supply and its adapters	1
5	Ethernet cable	1
6	USB A–male to micro B cable	1
7	microSD with adapter	1

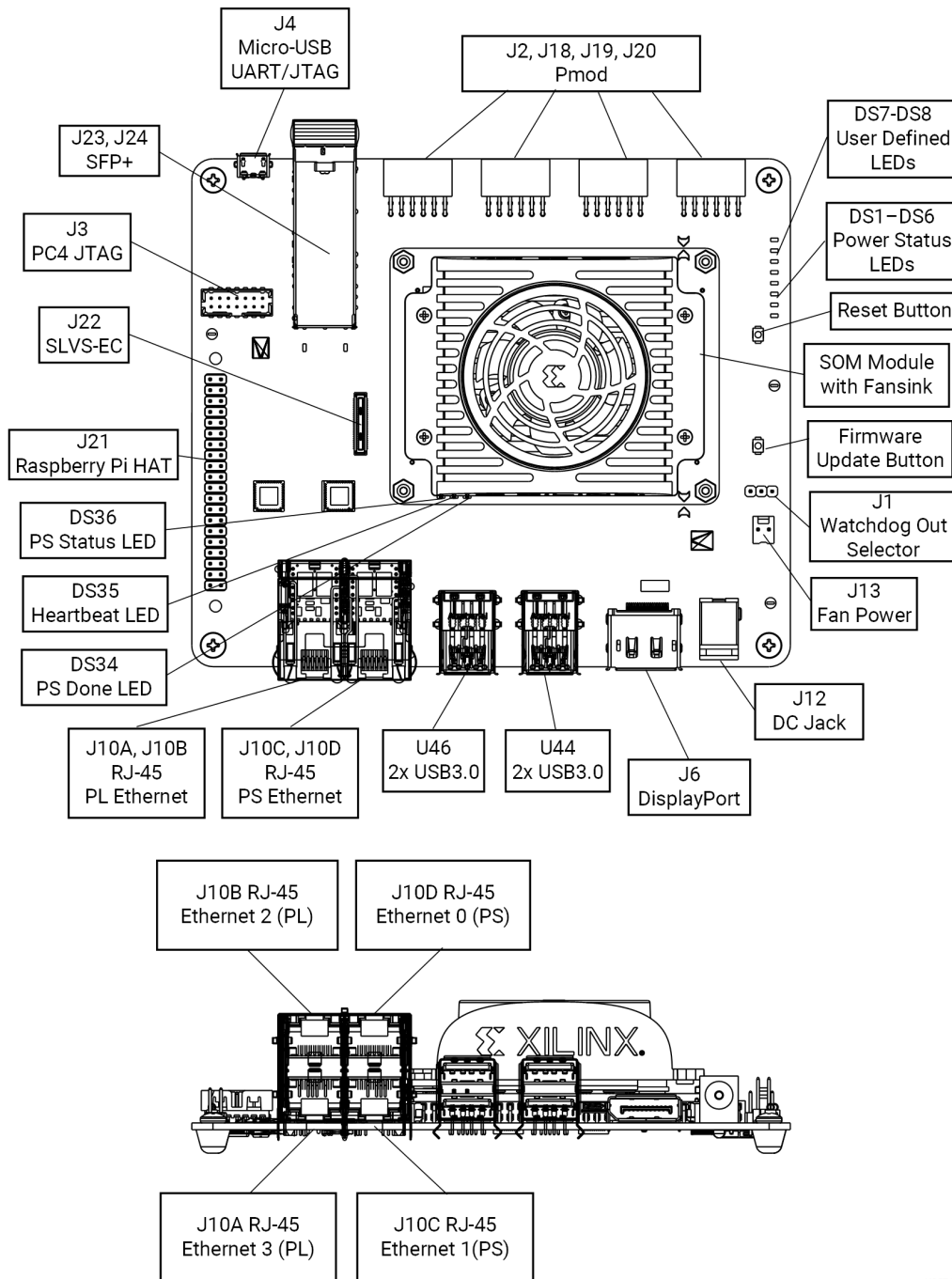
Interfaces

The following figure and table provide an overview of the physical connections, their designators, and relative position on the starter kit. The table uses the following abbreviations to indicate if a specific designator is located on the carrier card or on the SOM.

- CC = Device or interface is located on the carrier card
- SOM = Device or interface is located on the SOM

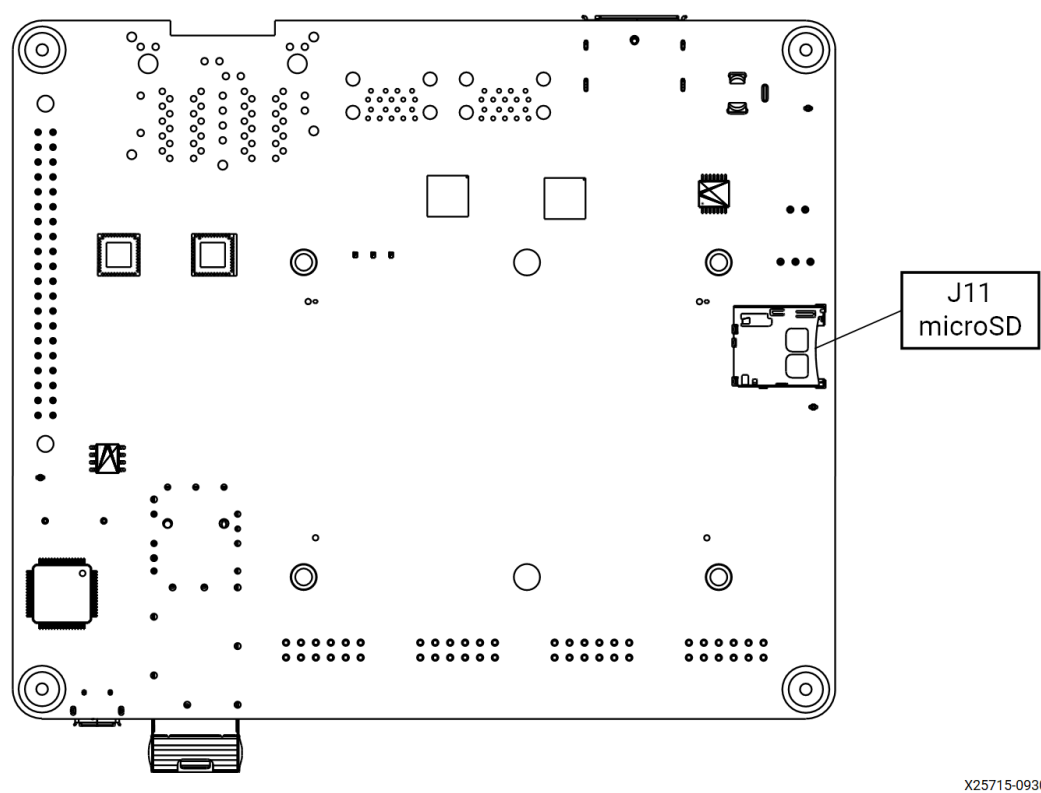
On the carrier card, there are four USB ports. USB0 and USB1 are each connected to a pair of USB physical port interfaces. There are four Ethernet interfaces with one pair connected to PS GEMs and one pair to PL–based Ethernet controllers. As shown in the following figure, Ethernet 1 (PS) on J10C is the default firmware and software Ethernet interface used for the image recovery application and the primary Ethernet interface in Linux.

Figure: Interfaces and Connectors—Top of Card



X25711-031124

Figure: Interfaces and Connectors—Bottom of Card



X25715-093021

Table: Descriptions and Locations

Location	Name	Description
SOM DS34	PS done LED	Lit indicates that the PS has successfully loaded a PL design.
SOM DS35	Heartbeat LED	Periodic flashing green LED driven by the Zynq UltraScale+ MPSoC APU processor.
SOM DS36	PS status LED	Status LED, when lit indicates a successful application load.

Location	Name	Description
CC DS1–DS6	Power status LEDs	Indicates various power supply and power domain status. Green LED indicates good status.
CC J2, J18, J19, J20	Pmod	Digilent Pmod 2x6 expansion header
CC J1	Watchdog out selector	J1 is used to select the external watchdog circuit. The default (pin 1–2) routes the output of the watchdog to the PMU GPI on MIO26. In the other position (pin 2–3) routes to the SRST pin of the Zynq UltraScale+ MPSoC.
CC J3	PC4 JTAG	Direct JTAG interface, bypasses the FTDI device.
CC J4	FTDI USB2.0 UART and JTAG	Integrated JTAG and device UART interface via USB2.0
CC J6	DisplayPort	DisplayPort video output
CC J10A	Ethernet RJ45 jack	1 Gb/s Ethernet 3 (PL) RGMII interface on HPB
CC J10B	Ethernet RJ45 jack	1 Gb/s Ethernet 2 (PL) RGMII interface on HPA

Location	Name	Description
CC J10C	Ethernet RJ45 jack	1 Gb/s Ethernet 1 (PS) RGMII interface
CC J10D	Ethernet RJ45 jack	1 Gb/s Ethernet 0 (PS) SGMII interface
CC J11	microSD card	microSD card boot device
CC J12	12V power input	12V power input jack
CC J13	Fan power	12V SOM fan power interface
CC J21	Raspberry Pi HAT	Raspberry Pi expansion header for HAT interface
CC J22	SLVS-EC	Framos FPA SLVS-EC interface
CC J23	SFP+	SFP+ connector
CC J24	SFP+ cage	SFP+ cage
CC SW1	Firmware update button	Push button used during the boot firmware update process
CC SW2	Reset button	Push button that resets the SOM via the device POR_B signal
CC U44	USB0	Two USB3.0 or USB2.0 compatible connectors
CC U46	USB1	Two USB3.0 or USB2.0 compatible connectors

Initial Setup

Powering the Starter Kit and Power Budgets

The AMD Kria™ KR260 Robotics Starter Kit requires a 12V, 3A power supply adapter to power the kit. The adapter is included in the kit. The power supply adapter barrel connector plugs into the DC jack (J12) to supply the +12V power source to KR260 Starter Kit.

Powering the K26 SOM

1. The KR260 Starter Kit carrier card on-board regulator generates a 5V supply and provides power to other voltage regulators.
2. The SOM power rail (V_{CC_SOM}) is powered by the 5V supply.
3. Next, the SOM on-board power-on sequencing starts.
4. The carrier card provides the programmable logic (PL) the V_{CCO} voltage rails after the SOM asserts the $V_{CCOEN_PS_M2C}$ and $V_{CCOEN_PL_M2C}$ signals.

Power Telemetry

A power monitor device is available on the SOM power rail (SOM_5V0). You can access the total power consumed by the SOM module through the I2C bus and AMD provided utilities.

Powering Peripherals

The KR260 Starter Kit carrier card supplies power to the I/O peripherals as specified by the following interface specifications.

USB3.0

There are four USB3.0 interface ports available on the KR260 Starter Kit carrier card. There are two independent USB controllers, but they share a power source. Each port can deliver a 5V supply to the attached I/O

peripherals with up to 900 mA per port. All ports are protected against an over-current event through 1.0A power switches per pair.

 **Note:** The total current (across all four ports) is allocated at 2.0A.

Pmod Connector

The 12-pin Pmod interface (from Digilent Inc.) is specified to be 3.3V, 100 mA. The four ports are supported by a 3.3V, 1.0A shared capacity across all connectors.

Raspberry Pi Expansion Header

The Raspberry Pi expansion header is for use with Raspberry Pi HATs. This 40-pin interface connector is supported by 3.3V and 5.0V supply voltages. There is a 1.0A limit per voltage rail.

SFP+

Pluggable SFP+ transceiver modules are supported by the SFP+ cage that provides a 3.3V, 600 mA supply budget. The total power consumed must fall within the power budget for the SFP+ module. The SFP+ power is not explicitly limited, thus care must be taken when attaching optional accessories to your carrier card.

Framos FPA SLVS-EC

The Framos FPA SLVS-EC connector is supported with two voltages. The carrier provides 1.8V at 800 mA and 3.8V with a limit of 1.0A. Because the 1.8V is not explicitly limited, care must be taken when attaching accessories to ensure a proper power budget.

microSD Card

The microSD card is supported by the 3.3V supply voltage with a power budget of 200 mA. This should be more than adequate for standard cards. It is also not explicitly limited. The starter kit supports up to 64 GB microSD cards.

Fan and Heat Sink

The KR260 Starter Kit is built with an integrated active cooling solution (see [Figure 1](#)). The integrated fansink allows you to exercise the full 10W AMD Zynq™ UltraScale+™ MPSoC application power budget without any additional accessories.

Out of the box, the 12V fan should already be plugged into the starter kit. If it is not, be sure that the fan is plugged into the connector designated in [Table 1](#). The fan connector is keyed to ensure proper orientation.

By default, the fan runs at a constant speed. Variable fan speed control can be implemented through a FPGA based PWM fan controller. The fan gating signal is connected to a FPGA HD I/O bank pin for control. Consult the corresponding KR260 Starter Kit carrier card schematic for specific pin assignments.

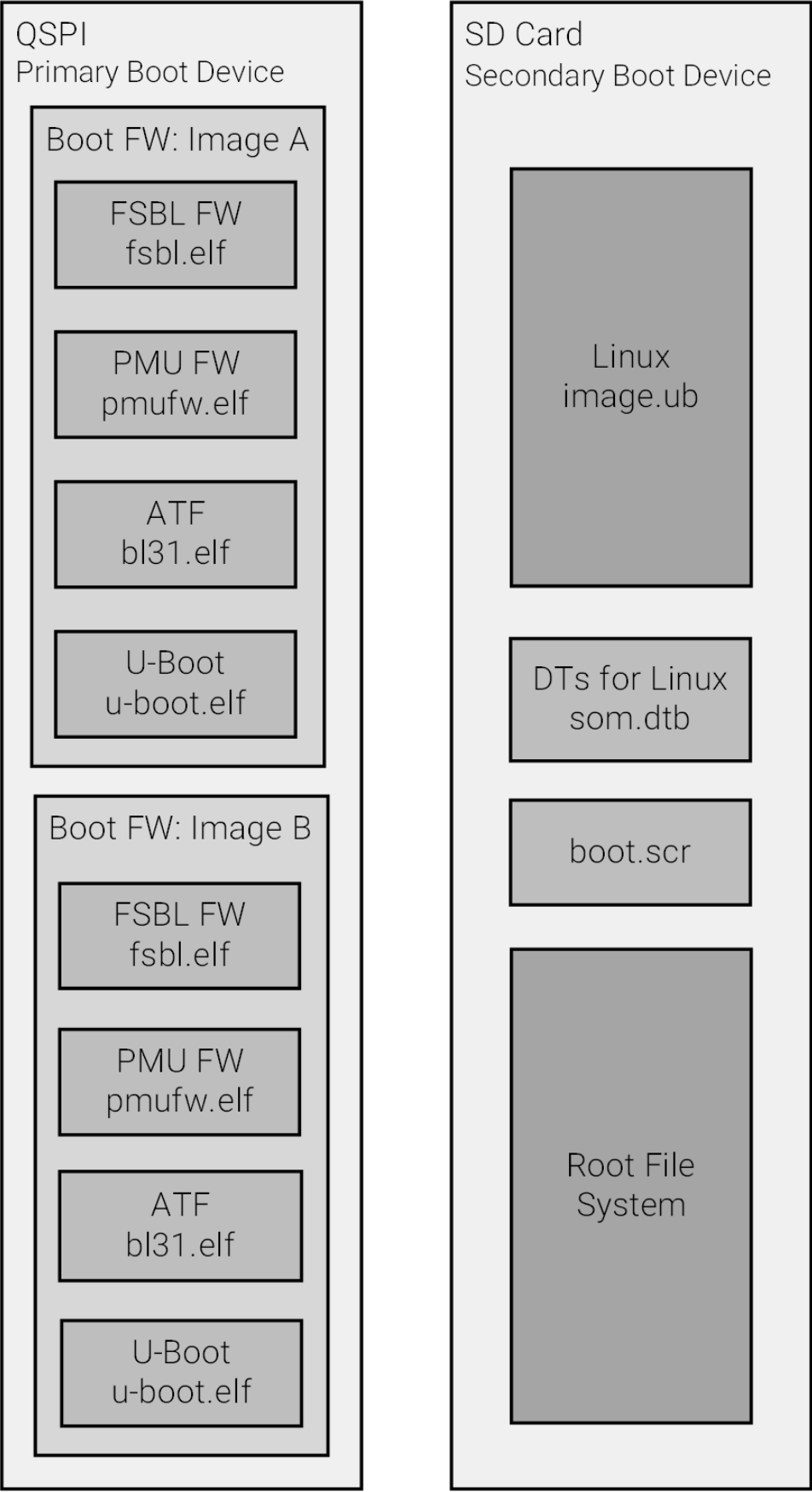
Boot Devices and Firmware Overview

The AMD Kria™ KR260 Robotics Starter Kit has a primary and secondary boot device that provides isolation of platform-specific boot firmware from the runtime operating system and application. This allows you to focus on developing and updating your application code within the application image without having to rebuild and flash boot firmware. The primary boot device is a QSPI memory located on the SOM and the secondary boot device is an SD card interface on the carrier card. By default, the KR260 Starter Kit carrier card sets the XCK26 boot mode to QSPI32. The SOM boots up to U-Boot using the QSPI contents and then U-Boot does a hand-off to the secondary boot device.

 **Note:** You must flash the SD card image and populate the microSD card in the carrier card for the kit to successfully boot to Linux.

The overall boot device definition and firmware contents are outlined in the following figure.

Figure: Boot Devices



X26536-051122

Important: Production SOMs provide both QSPI and eMMC devices on the SOM PCB to support integrated primary and secondary boot configurations.

Primary Boot Device

The primary boot device is a QSPI device located on the SOM. The necessary elements are packaged in an AMD Zynq™ UltraScale+™ MPSoC specific format and file captured as BOOT.BIN. The BOOT.BIN file contains the board-specific boot firmware that consists of the following elements:

- **FSBL**
First-stage boot-loader firmware
- **PMU**
Platform management unit firmware
- **ATF**
Arm® trusted firmware
- **U-Boot**
Second-stage boot loader

U-Boot provides the functionality for the hand-off between the primary boot device and the secondary boot device. It searches through a U-Boot defined prioritized list of secondary boot devices with priority given to the SD card interface.

The primary boot device provides a redundant copy of boot firmware arranged in an A/B configuration. The A/B configuration provides a dynamic primary and secondary image operation with corresponding update mechanisms. On boot, the system automatically boots from the defined primary image, and, if boot fails, it falls back to the previously known good boot image.

Secondary Boot Device

The secondary boot device on the KR260 Starter Kit is the SD card. It contains the operating system image and associated application files. The KR260 Starter Kit accelerated application references are built on the Linux operating system. The [Getting Started](#) webpage provides a pre-built

reference image that can be written to a microSD card for out-of-the-box functionality. SOM board support packages (BSPs) are also available if you want to customize your OS.

 **Recommended:** The SOM is designed to use SDHC standard microSD cards. See [AR66779](#) for a list of tested microSD cards.

Software Getting Started

To get started with the KR260 Starter Kit, prior to powering, booting the board, and running your first application, you need to follow the instructions on the [Getting Started with Kria KR260 Robotic Starter Kit](#) pages to download and write the Xilinx SOM Starter Linux image to a microSD card. The webpage guides you to power on the KR260, boot Linux, and run a number of pre-built accelerated applications to start evaluation of the capabilities on the platform.

Platform Management Utility

A platform management utility called `xmutil` is included in the SOM Linux image to help you configure and work with the SOM. Refer to the [Kria SOM Wiki page](#) for a list of `xmutil` functions and their descriptions.

Accelerated Applications

The AMD SOM platforms are enabled with a number of accelerated applications (AA) that can be dynamically installed on the SOM platform. The SOM starter Linux image is application agnostic and provides a set of utilities for pulling the hardware accelerated application examples from the SOM Linux package feeds.

Accelerated applications are software controllable, application-specific reference designs for roboticist, AI developers, embedded developers, and system architects to customize and enhance the functionality through software control or updating the AI models.

The AMD Kria™ robotics stack, Kria Robotics Stack (KRS) Tutorial ([XD042](#)) is an integrated set of robot libraries and utilities that use hardware to accelerate the development, maintenance, and commercialization of industrial-grade robotic solutions. It adopts the robot operating system

(ROS) Software Development Kit (SDK) and enables a ROS 2–centric development approach that spans from the creation of computational graphs to the commercialization of ROS 2 overlay workspaces found in the Kria [App Store](#). Some robotic accelerated applications are developed using the KRS.

The following table outlines some featured accelerated applications available for the KR260 Starter Kit. Consider this list as a starting point. Visit the SOM [Getting Started](#) webpage for the most up–to–date accelerated applications availability.

Table: KR260 Accelerated Applications Overview

Name	Description
ROS 2 Perception Node	Using the hardware accelerated perception for ROS 2 developers makes it easier to build high–performance solutions on Kria platforms. The focus of this accelerated application is to improve the throughput of image processing, an important facet to roboticists leveraging the Kria robotic stack (KRS) and Vitis Vision Library . The accelerated application reduces the load on the host CPU while providing significant performance gain. The data source and virtual environment are provided by the Gazebo simulator with Ubuntu Linux 22.04.

Name	Description
ROS 2 Multi-Node Communications via TSN	The ROS 2 multi-node communication via TSN is an accelerated application that focuses on applying ROS 2 within the context of an time-sensitive networking (TSN) based communications infrastructure that is developed using the Kria robotics stack.
10GigE Vision Camera	The 10GigE vision camera is a hardware accelerated machine vision application that is used in defect detection on the KR260 platform. It uses the Framos SLVS-EC sensor interface, the Euresys 10GigE vision interface, and the defect detection algorithm using the Vitis Vision Library .

Accelerated Application Package Selection

 **Recommended:** Public Ethernet connectivity is necessary to dynamically pull down the latest accelerated application designs.

1. If you have not already verified Internet connectivity do so before proceeding via ping test or DNS lookup (e.g., nslookup).

2. Example applications are deployed using the Linux package management framework for over-the-air deployment. The `sudo xmutil getpkgs` lists a series of package groups that can be installed on your platform. The package group naming convention is: `packagegroup-kit_name-application_name`. For example, the machine vision application for the KR260 platform has the following package group name `packagegroup-kr260-machine-vision`. You can install any number of matching accelerated applications to your platform. For exact commands to install, refer to the [Kria SOM Wiki](#) for further details.

 **Note:** You should only install package-groups that are compatible with your particular starter kit configuration.

3. For any applications installed on the local file system via the package feed, the platform can now dynamically load and swap those applications. To see a list of the applications local to the system, execute `sudo xmutil listapps`. You can also see what applications are local by manually exploring the `/opt/xilinx` directory.
4. By default, `kr260-dp` is loaded on boot. From the applications list, check for an active application loaded (active = 1 in the `xmutil listapps` output). If there is a loaded application, unload it by running the `sudo xmutil unloadapp` command to unload the current application before proceeding to the next step.
5. From the application list, start the new application by running `sudo xmutil loadapp application_name`. The platform configuration is automatically handled and starts the application.
6. Applications with a Jupyter-based cockpit will start-up automatically. You need to point your web-browser to the associated IP address and port. The associated IP address and port information is printed to the UART at boot. To query your Jupyter lab server URL after the initial boot, run: `sudo jupyter notebook list`.

Supported Peripherals

The following table outlines external peripherals that are tested with the corresponding accelerated applications. It is recommended that you use a peripheral from the list to ensure that you realize maximum platform performance.

Table: Accelerated Application Peripherals

Accelerated Application	Peripheral	Part Number
10GigE Vision Camera	Sony IMX547 camera kit color for SLVS_EC (J22)	Sony HW-IMX547C-SK-G
	Sony IMX547 camera kit monochrome for SLVS_EC (J22)	Sony HW-IMX547M-SK-G
	10GBase-SR SFP+ transceiver (J23)	10Gtek AXS85-192-M3

The following table outlines external peripherals that are functionally verified with the KR260 Starter Kit.

Table: KR260 Starter Kit Functionally Tested Peripherals

Peripheral	Part Number
Pmod RS485 isolated communications	Digilent 410-310
USB 4K camera	Logitech BRIO

For a complete list of KR260 compatible accessories, refer to the [Kria SOM Forum](#).

Tools Integration

The K26 SOM and KR260 Starter Kit are integrated with the AMD Vitis™ software development platform and AMD Vivado™ Design Suite for rapid development of your unique applications on the platform.

Vivado Board Flow

The K26 SOM is enabled in the Vivado Design Suite through the Vivado Board Flow functionality. Vivado Board Flow enables a level of hardware

abstraction that automatically configures peripherals fixed on the SOM card (e.g., DDR4), defines associated timing constraints, and presents the customizable physical I/O available on the SOM connector(s).

The Vivado SOM board model is available through the Vivado installation process as well as on the Vivado board file [GitHub](#) repository. The following KR260 related Vivado board files are available.

- **KR260 Starter Kit**
Configured K26 SOM with Robotics Starter Kit companion card
- **SM-K26-XCL2GC**
K26 commercial grade production SOM
- **SM-K26-XCL2GI**
K26 industrial grade production SOM

The AMD SOM board flow infrastructure provides starter kit carrier card awareness through the Vivado tools companion card mechanism. Automation for I/O connection and peripheral IP configuration when selecting a SOM and an associated carrier card, such as the KR260 Starter Kit, is used to create a hardware configuration.

For additional information on using the Vivado tools and the Vivado board flow, refer to the Vivado Design Suite User Guide: System-Level Design Entry ([UG895](#)).

Board Reset, Firmware Update, and Recovery

This section outlines the update and recovery mechanisms built into the KR260 Starter Kit. Two tools are provided for firmware updates. The first is a Linux based A/B update tool that supports remote and redundant firmware updates to the A/B firmware partitions of the QSPI device with custom or AMD provided updates. The second tool is the Ethernet recovery tool that is intended to be used only when recovering a full platform to the original factory firmware.

Firmware Update

The firmware update button is the physical SW1 /FWUEN push button located on the KR260 Starter Kit carrier card. The button is used to support

two features:

1. To force the platform into a recovery mode application during power-on. The recovery application is described in the [Boot Image Recovery Tool](#) section of this document.
2. Security mechanism to prevent remote update of the boot firmware, without the user being physically present.

Boot Image Recovery Tool

The boot image recovery tool is a small application included in the AMD provided Starter Kit QSPI image. It provides a simple Ethernet-based interface and application for updating the boot firmware if Linux is not functional. If Linux is functional, the recommendation is to update the boot firmware using the `xmutil` boot firmware update utilities. Refer to the [Kria SOM Wiki page](#) for details on how to use the boot image recovery tool.

Boot Firmware A/B Update

As outlined in the [Boot Devices and Firmware Overview](#) section, the AMD starter kit and SOM provide two copies of the boot firmware in the QSPI device. This mechanism has a robust update infrastructure through a ping-pong methodology, where the last known good boot image is always kept available in the platform. The AMD starter kit Linux provides a utility for doing these updates entirely on-target. The steps for using the AMD SOM A/B update tool are documented in the Wiki page at: [Boot FW Update Process](#).

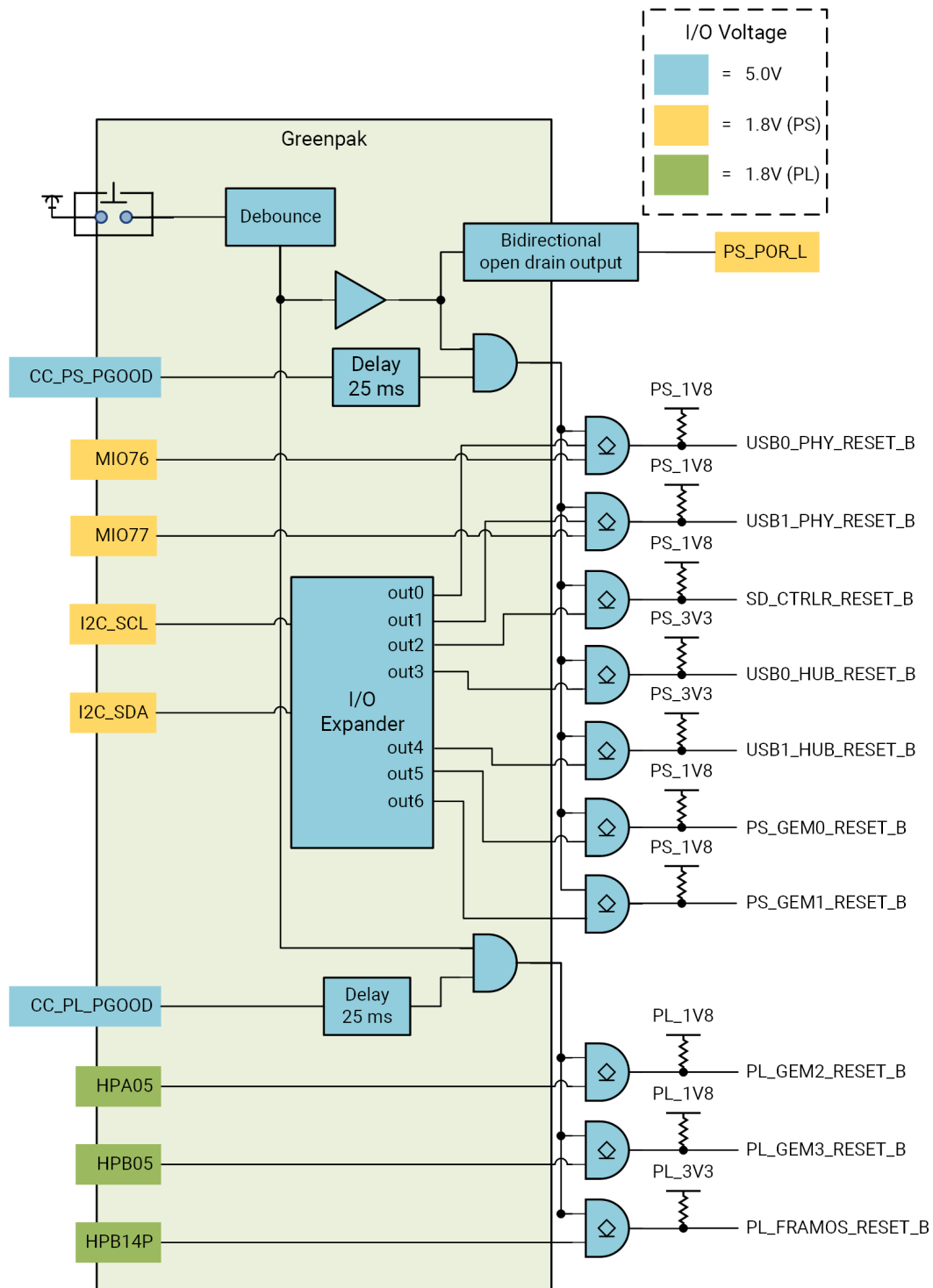
Board Reset

Power-on Reset

1. The SOM reset signal (PS_POR_L) is held in reset until the CC_PS_PGOOD signal is asserted on the carrier card.
2. All the PS and PL I/O device reset signals on the carrier card are held in reset until 25 ms after the PS and PL power domain are powered up and stable.
3. You can perform a hard reset on the KR260 Starter Kit by pressing the reset button (CC SW2) or by commanding a reset through software.

The following figure shows the reset functions supported on the KR260 Starter Kit.

Figure: Device Reset



X25670-082421

Additional Resources and Legal Notices

Finding Additional Documentation

Technical Information Portal

The AMD Technical Information Portal is an online tool that provides robust search and navigation for documentation using your web browser. To access the Technical Information Portal, go to <https://docs.amd.com>.

Documentation Navigator

Documentation Navigator (DocNav) is an installed tool that provides access to AMD Adaptive Computing documents, videos, and support resources, which you can filter and search to find information. To open DocNav:

- From the AMD Vivado™ IDE, select Help > Documentation and Tutorials.
- On Windows, click the Start button and select Xilinx Design Tools > DocNav.
- At the Linux command prompt, enter `docnav`.

 **Note:** For more information on DocNav, refer to the Documentation Navigator User Guide ([UG968](#)).

Design Hubs

AMD Design Hubs provide links to documentation organized by design tasks and other topics, which you can use to learn key concepts and address frequently asked questions. To access the Design Hubs:

- In DocNav, click the Design Hubs View tab.
- Go to the [Design Hubs](#) web page.

Support Resources

For support resources such as Answers, Documentation, Downloads, and Forums, see [Support](#).

References

These documents provide supplemental material useful with this guide:

1. [Kria SOM GitHub.io documentation](#)
2. Kria KR260 Robotics Starter Kit Data Sheet ([DS988](#))
3. Kria K26 SOM Data Sheet ([DS987](#))
4. Kria SOM Carrier Card Design Guide ([UG1091](#))
5. Vitis Model Composer User Guide ([UG1483](#))
6. Vitis Unified Software Platform Documentation: Application Acceleration Development ([UG1393](#))

Revision History

The following table shows the revision history for this document.

Section	Revision Summary
4/24/2024 Version 1.1	
Interfaces	Added J1 description to table. Clarifying edits to the Ethernet tags on Figure 1 .
Platform Management Utility	Removed details and referred to Kria SOM Wiki.
Accelerated Applications	Updated links.
Boot Image Recovery Tool	Renamed Ethernet recovery tool to boot image recovery tool. Removed Ethernet Recovery Tool Interface image and details, and referred to Kria SOM Wiki.
Boot Firmware A/B Update	Removed details and referred to Kria SOM Wiki.
5/17/2022 Version 1.0	
Initial release.	N/A

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